# Worksheet for Bayesian Optimization

1. Assume that
$$f \sim GP\left(\mu(x), k(x, x')\right)$$
and that
$$y_i \mid f(x_i) \sim N\left(f(x_i), \sigma^2\right)$$
What is the distribution of $f \mid y_i$? This is well defined even if $\sigma^2 = 0$.

   Hint1: This will be another Gaussian process, and is thus characterized by the conditional mean
$$\mathbb{E}\left[f(x) \mid y_i\right]$$
and the conditional covariance
$$\mathbb{E}\left[f(x)f(x') \mid y_i\right] - \mathbb{E}\left[f(x) \mid y_i\right] \mathbb{E}\left[f(x') \mid y_i\right]$$

   Hint2: You may want to consider the joint (unconditional distribution) of $(y_i, f(x))$ and $(y_i, f(x), f(x'))$.

2. How can you represent this in code using the kernel function? If we're doing Bayesian optimization, we may be evaluating the posterior mean and posterior covariance many times for each update.

3. How does this change if we condition on multiple observations? Suppose that we've already conditioned on a set of $n$ observations $\{(x_i, y_i)\}_{i=1}^{n}$. How can we condition on one more?

4. What are the gradients of the posterior mean and variance with respect to $x$? That is, we assume that
$$f_{post} \sim GP\left(\mu_{post}(x), k_{post}(x, y)\right)$$
In terms of the representation that you found in 2., what is $\nabla_x \mu_{post}(x)$, and $\nabla_x k_{post}(x, x)$?

**Setup for a PDE inverse problem:** Suppose that $u(x,t)$ satisfies the PDE

$$
\begin{aligned}
u_t &= \alpha u u_x + \kappa u_{xx} && (1)\\
u(x,0) &= a\sin(6\pi x) - b\cos(2\pi x) - 1 && (2)\\
u(0,t) &= u(1,t) && (3)\\
& && (4)
\end{aligned}
$$

with for some parameters $(\alpha, \log(\kappa), a, b)$.

We observe points $\{(x_i, t_i, u_i)\}_{i=1}^n$ where $u_i = u(x_i, t_i)$. Your goal is to recover the parameters $p = (\alpha, \log(\kappa), a, b)$ from data, where we assume that

$$
\begin{aligned}
\alpha &\in (0,1) && (5)\\
\log(\kappa) &\in (-6,0) && (6)\\
a &\in (-5,5) && (7)\\
b &\in (-5,5) && (8)\\
& && (9)
\end{aligned}
$$

We've provided code that solves the PDE above for any set of parameters $p$, giving us a solution $u_p(x,t)$ for any $p$ (but evaluating for any value of $p$ requires a PDE solve).

Try to recover the true parameters $p^*$ by finding parameters $\hat{p}$ to minimize

$$
L(p) = \frac{1}{2}\sum_{i=1}^{n}(u(x_i,t_i) - u_i)^2 \qquad (10)
$$

using Bayesian optimization.

We've provided code for both this solution map $p \to u_p$, and which evaluates the loss $L(p)$.